

TECHNIQUES

- Modules $(M, R, +)$

↳ Verify the axioms

- Submodules

↳ Verify the axioms (Check subgroup and
for any $r \in R, m \in M$
 $rm \in M$)

- Homomorphism

↳ Verify the axioms

Some Non Trivial / Unobvious examples.

EXAMPLES

Modules

- Vector space (Ring over module is a field).
- (V, T) where V is a vector space and T is a linear transform. Ring is constructed to be $\mathbb{F}[x]$ where

$$f(x) \cdot v := f(T)(v)$$

Note: $(V, T) \longleftrightarrow \mathbb{F}[x]$ module.

For $\longrightarrow$ we define $T(x) := x \cdot v$

Submodules

- Vector subspaces
- In $\mathbb{Z}$ -modules say M
 N is a submodule iff $(N, +) < (M, +)$.
- In R is a R -module
 N is a submodule iff N is an ideal.
- If M is a $\mathbb{F}[x]$ -module $\cong (V, T)$
 W is a submodule iff $T(W) \subseteq W$
i.e. W is invariant under T .

Module	f.g.	cyclic
$(V, \mathbb{F})$	$\dim V < +\infty$	$\dim V = \{0, 1\}$
G is abelian group treated like $\mathbb{Z}$ -module	f.g. as group	cyclic as group
R comm ring and $M \triangleleft R$ M is R -module	f.g. as an ideal	principal ideal

Module homomorphisms

◦ Abelian groups $\longleftrightarrow$ $\mathbb{Z}$ -modules

then group homo $\longleftrightarrow$ module homo^m

◦ For vector fields

linear transformation $\longleftrightarrow$ module homo^m

◦ For ring R , $\theta: M \longrightarrow N$ where
 $M = N = R$

$$\theta(rs) = r\theta(s)$$

is NOT a ring homo^m but is a module
homo^m.

* Isomorphism theorems hold for modules.

